

Product Requirements Document

Benz Auto Workshop Management System

Project	P02 — Workshop Management System
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1. About This Document

The signed quotation lists **what** Zedech will deliver. This document explains **how** each piece will work and **when** it will be delivered, so that both sides walk into the next four months with a shared picture.

This is a **discussion document**, not a contract — the quotation already covers the commercial terms. The intent here is simply for KL and the Zedech team to read the same description of the product and agree it matches what was discussed.

If anything in this document doesn't match how you (KL) want the project to run, please flag it and we'll adjust.

How to read this document:

- Section 2 shows the full delivery roadmap.
- Section 3 explains the two core ideas the system is built around (Appointment, Task).
- Sections 4–6 walk through every module, phase by phase.
- Section 7 covers maintenance, hosting, and ongoing usage costs.
- Section 8 covers how Zedech and Benz Auto will work together over the four months.

2. Roadmap at a Glance

The 4-month engagement is split into three delivery phases that map to the payment milestones in your signed quotation.

Phase	Weeks	Payment milestone	Theme
Kickstart	(signed)	20% deposit — paid this week	Project starts; Meta/WhatsApp Business approval initiated
Phase 1	8 weeks (Weeks 1–8)	30% on Phase 1 acceptance	Customer engine: Chatbot, CRM, Unified Inbox, core workshop tools
Phase 2	4 weeks (Weeks 9–12)	25% on Phase 2 acceptance	Workshop operations: Outreach, Suppliers, AutoCount, Analytics
Phase 3	4 weeks (Weeks 13–16)	25% on Phase 3 acceptance	OpenClaw AI platform, Agents, Website, final tuning

What you get at each milestone

End of Phase 1 (Week 8):

A working WhatsApp chatbot that already speaks English, Chinese, and Malay (the customer-facing surface is multilingual from launch). A unified inbox for all customer messages. A customer database with vehicle history filed by plate. A digital Workshop Board (kanban for technician tasks) and Office Board (kanban for admin tasks) — both fully customizable. An appointment calendar (day, week, month views). Photo capture for technicians, auto-filed to the vehicle. Three fixed roles (Superadmin, Admin, Technician). A walk-in onboarding form (also trilingual). Basic outstanding-payment tracking. Outreach templates ready to blast — manual campaigns to customer segments, plus auto review-requests when an appointment is released.

End of Phase 2 (Week 12):

Enforced workflow compliance for technicians. Editable roles with per-page/per-action permission gating (each workshop can shape roles to its own structure). Bay-centric calendar view. Automated maintenance reminders and pre-appointment booking reminders. Supplier order tracking with AI assistance. AutoCount stock and pricing visibility. Sales and margin analytics with RFM segmentation feeding back into outreach campaigns. Facebook Messenger, Instagram DM, and TikTok DM channels added to the inbox. Full V2 polish on Phase 1 modules (auto-assignment, dormant flagging, no-show automation, urgency flags, multiple boards, saved views, etc.).

End of Phase 3 (Week 16):

OpenClaw AI platform installed on your private server. Two AI agents live (Social Media & Content, Pricing Strategy). The public Benz Auto website online. Daily morning brief delivered to your inbox. Final tuning week for adjustments based on real-world usage.

3. Two Core Ideas

The whole system is built around two simple ideas. Locking these in now prevents confusion later.

Term	What it means	Example
Appointment	A booking on the calendar — your customer's commitment to come in. Tied to a customer and vehicle. Has a time slot and a bay reservation.	"Mr Tan, Civic ABC1234, Saturday 10am, Bay 3"
Task	An internal unit of work for your staff. Carries a label saying who it's for: admin (front desk / sales) or technician (workshop floor).	Admin task: "Order brake pads from Lim Auto", "Call Mr Tan to confirm tomorrow". Technician task: "Replace brake pads — Civic ABC1234", "Oil change — Saga ABC5678".

That's it. Customers and Vehicles are obvious record types. Everything else in the system is a view, filter, or report on top of these two ideas.

How an Appointment moves

An Appointment goes through three simple states:

Booked → Arrived → Released

- **Booked** — customer agreed to come at a specific time
- **Arrived** — car is in the workshop, work is happening
- **Released** — car has left the workshop, the visit is closed

When a customer books on WhatsApp, the appointment is created in **Booked**. When the car shows up at the workshop, it's switched to **Arrived** and its associated technician tasks become live. When the customer drives off, **Released**, and review requests / invoicing kick in.

Task boards — fully customizable, Notion-style

Both task types live on **fully customizable kanban boards** that you (KL) shape however you want. Think of it like Notion: rename columns, reorder them, add new ones, delete them, color-code them — at any time.

Two boards, independent of each other:

- **Workshop Board** (technician tasks) — e.g., your default could be
Pending → In Progress → Completed , but you might prefer
Diagnose → Awaiting Parts → Wrenching → QC → Test Drive → Done . Up to you.
- **Office Board** (admin tasks) — e.g., default To Do → Done , but you could turn it into
Inbox → This Week → Waiting on Customer → Done . Up to you.

When the car arrives for an appointment, **one technician task is created per service item** (one for oil change, one for brake pads, etc.) and they all enter the Workshop Board's first column.

One small rule per board: mark the "Done" column

So that the system knows when a task is truly finished, you flag **one column on each board as the "Done" column**. That's the only constraint on board customization. The system uses the Done flag to:

1. Count completed tasks correctly in analytics
2. Trigger downstream actions — most importantly, when *all* technician tasks for an appointment land in Done, the appointment is eligible to flip to Released (which fires the review request)

That's it. Everything else about the boards is yours to shape.

Workflow rules attach to transitions you define

In Phase 2 (the Workflow Compliance module), you can attach mandatory steps to *any specific column-to-column transition* you choose. For example: "moving a card into Test Drive requires a photo upload" or "moving into Done requires customer sign-off." You pick the transition, you pick the required steps.

What you see on the dashboard

The three counters at the top of the screen pull from the model above:

- **Cars currently in workshop** — appointments in state **Arrived**
- **Appointments today** — appointments scheduled for today (any state)
- **Pending tasks** — all tasks (admin + technician) not yet Done / Completed

4. Phase 1 — Customer Engine (Weeks 1–8)

Phase 1 delivers the customer-facing engine that drives daily operations: the chatbot answering messages, the unified inbox where staff reply, the CRM holding customer and vehicle records, and the workshop tools that turn appointments into completed work.

A note on languages: the **chatbot, the digital onboarding form, and any other customer-facing surfaces speak English, Chinese, and Malay from Day 1**. The **staff-facing admin UI is in English** — your team works in it daily and English is standard for workshop management software in Malaysia. (If translation of the admin UI becomes important later, it can be added as a separate quotation.)

Each module below is described in three layers:

- **MVP** — what ships in Phase 1
- **Future expansion** — features that arrive in Phase 2 or 3 (already paid for, just sequenced later)
- **Defaults** — choices Zedech has set on your behalf for speed; you can override any of these at any time

4.1 AI Chatbot (Concierge)

Purpose: A 24/7 virtual assistant on your WhatsApp number that handles the most common customer messages so your staff can focus on the harder ones.

MVP (Phase 1):

- Replies in **English, Chinese, and Malay** — auto-detects the language of each incoming message
- Replies within 1 minute, 24/7

- Answers common questions about hours, services, and pricing from a knowledge base you can update yourself (no coding)
- Auto-categorizes incoming messages: booking, quote, complaint, general inquiry, spam
- Pulls details like license plate, car model, and dates out of customer messages and pre-fills records for staff
- One-click handoff to a human staff member from inside the unified inbox
- Politely rejects spam and archives those conversations
- Live monitoring view: see what the AI is replying in real time
- AI toggle: turn the bot off for specific customers
- Master switch: turn the bot off across the whole system
- Customers can fully self-serve booking through WhatsApp without staff involvement
- Auto-creates an Appointment when a booking is confirmed (and spawns matching technician tasks once the car arrives)
- Handles confirm / cancel / reschedule flows end-to-end autonomously

Future expansion (Phase 2 onwards):

- Auto-follows-up with stalled inquiries and tags them "Potential Client"
- Facebook Messenger, Tik Tok and Instagram DM channels
- Records suspicious activity (jailbreak attempts) for review

Defaults Zedech has set:

- Spam policy: the bot dismisses politely and archives. Spam never reaches the inbox.
- Knowledge base format: structured admin forms (operating hours, service prices, FAQ entries) — not raw JSON.

4.2 Customer Relationship Management (Module 2)

Purpose: A central place for every customer's contact details, vehicles, and full repair history.

MVP (Phase 1):

- ***Customer profile: name, phone, email, preferred language, address, notes, tags, e invoice detail***
- ***Multiple vehicles per customer (plate, make, model, year, color, engine, mileage, insurance expiry, road tax expiry, service interval, service and repair history, last pic, last technician)***
- Instant search by license plate, phone, or name (results appear as you type)

- Full service history per vehicle: every past appointment, every completed technician task, every photo
- Total spending per vehicle and per customer
- Each customer can be linked to a specific sales staff member

Future expansion (Phase 2 onwards):

- Auto-assignment rules for new inquiries (round-robin, least-loaded, by language, by channel)
- Dormant customer flagging — automatic detection of customers who haven't visited in a configurable period

Defaults Zedech has set:

- ***Vehicle field list: plate, make, model, year, color, engine type, VIN, insurance expiry, road tax expiry, service interval (km), last mileage.***
- Dormant threshold: 6 months since last visit (configurable from the admin panel from Phase 2 onwards).
- Each customer is assigned to one sales staff. If the staff member changes, history stays with the customer record.

4.3 Unified Inbox & Multi-Channel Messaging (Module 4)

Purpose: One inbox for every customer message across WhatsApp, Facebook, and Instagram, so nothing falls through the cracks.

MVP (Phase 1):

- WhatsApp Business channel fully integrated
- *Three-panel layout on desktop:*
 - Left: filter groups (Enquiry, Booked, In-Progress, plus any custom groups you create)
 - Middle: the conversation
 - Right: customer info, vehicle, linked appointment / tasks
- Send and receive photos, PDFs, and voice notes
- Predefined quick replies in English, Chinese, and Malay
- Attach a chat or specific messages to an Appointment or Task — visible from both the inbox and the linked record
- Mobile-friendly layout for phone use

Future expansion (Phase 2 onwards):

- Facebook Messenger channel
- Instagram DM channel
- Tiktok DM channel
- Separate "Suppliers" channel filter
- Advanced custom filter groups based on tags, status, channel, or assignment

Defaults Zedech has set:

- ***Default groups out of the box: Enquiry · Booked · In-Progress.***
- Conversations cannot be deleted, only archived (so audit trail is preserved).

4.4 Operations Dashboard & Workflow (Module 1)

Purpose: The home screen — at-a-glance visibility into what's happening in the workshop right now.

MVP (Phase 1):

- Live counters at the top: Cars in workshop · Appointments today · Pending tasks
- **Workshop Board** — drag-and-drop kanban for technician tasks. Columns are fully customizable by admin (rename, reorder, add, remove, color-code). Cards show: service item, plate, customer name, assigned technician, linked appointment time
- **Office Board** — drag-and-drop kanban for admin tasks. Columns also fully customizable. Cards show: title, assignee, due date, linked customer/appointment if any
- Each board has one column flagged as the "Done" column (the only structural constraint)
- Tabbed dashboard for admin: switch between Workshop Board and Office Board
- Technicians default to Workshop Board, only see their own tasks
- Mobile-friendly view of both boards (basic responsive layout — drag-drop polish on phones lands in Phase 2)
- Booking conflict prevention: the system blocks double-booking the same bay or staff at the same time

Future expansion (Phase 2 onwards):

- Urgency flagging with predefined delay reasons (admin-configurable dropdown) and free-text notes
- Real-time auto-refresh (counters update without page reload)
- Touch-optimized drag-and-drop on phones for technicians
- Multiple Workshop Boards (e.g., one per bay, or one per technician team)

- Saved board views with filters

Defaults Zedech has set:

- Workshop Board ships with Pending → In Progress → Completed (Completed flagged as Done). Customizable by you.
- Office Board ships with To Do → Done (Done flagged as Done). Can expand to a multi-column kanban any time.
- Booking conflicts are a hard block in Phase 1 — no manual override.

4.5 Appointment & Calendar System (Module 3)

Purpose: Schedule appointments without overbookings, see workshop utilization at a glance.

MVP (Phase 1):

- Day, week, and month calendar views
- Click an empty time slot to create a booking — pick customer, vehicle, intended services, bay
- Color-coded by appointment status (Booked / Arrived / Released)
- Hard block on double-booking the same bay at the same time
- Customer-self-service booking via chatbot

Future expansion (Phase 2 onwards):

- Filter by bay (technicians work tasks, not appointments)
- Bay-centric Gantt view (bays as rows, time as columns) — for at-a-glance workshop utilization
- One-way sync to Google Calendar (so staff see workshop schedule from their phones)
- Auto-highlight Malaysian and Johor public holidays
- No-show automation — when a customer doesn't show up, send WhatsApp follow-up + create staff task + flag the profile
- Holiday data can be edited.

Defaults Zedech has set:

- No-show automation policy: send a polite WhatsApp follow-up, create a "follow up with no-show" task for the assigned staff, and add a soft flag to the customer profile (does *not* block future bookings unless you turn that on).

4.6 Media, Documents & Vehicle History (Module 9)

Purpose: Every photo, document, and chat conversation is filed under the **vehicle** (indexed by license plate) — not the task, not the appointment, not the customer. Open any vehicle record and see its complete history at a glance.

Why vehicle, not customer? Many customers own multiple cars. Filing photos under "Mr Tan" alone makes it impossible to tell whether a brake-pad photo was for his Civic or his Hilux. Filing under the **vehicle** (by plate) eliminates that ambiguity.

MVP (Phase 1):

- In-app camera access via phone browser — technicians snap photos directly while working
- Photos automatically tagged with the vehicle plate and filed to that vehicle's record
- Upload from phone gallery — same auto-filing
- Each vehicle profile has a single chronological **history timeline** showing: photos, attached documents, linked chat conversations, completed technician tasks, past appointments — the full audit trail in one view
- PDF viewer inside the app (insurance letters, invoices, road tax screenshots, etc.) — also filed under the vehicle
- Replaces the current Telegram workflow for photo sharing

Future expansion (Phase 2 onwards):

- WhatsApp customer photos auto-saved to the relevant vehicle — when the customer has multiple vehicles, the chatbot asks "*Which car is this for?*" and tags accordingly. If the chatbot is unsure, it asks a staff member before filing.
- Configurable retention policy with admin warning before deletion
- Bulk export of a vehicle's full history (PDF report)

Defaults Zedech has set:

- **Vehicle (by plate) is the primary filing key for everything visual and textual.** A customer profile shows the list of their vehicles; clicking into a vehicle reveals its full history timeline.
- When a multi-vehicle customer sends media or messages, the chatbot disambiguates ("which car?") before filing. Single-vehicle customers skip this step.
- Default retention: photos kept indefinitely in Phase 1. Retention rules turn on in Phase 2 once you decide a comfortable timeframe.

4.7 Access Control & Audit (Module 10)

Purpose: Make sure each staff member sees only what they need, and every important action is logged.

MVP (Phase 1) — three fixed roles:

- **Superadmin** — Zedech and KL. Full system access including configuration, integrations, billing, role management.
- **Admin** — KL's office staff (front desk, sales). Full operational access: customers, vehicles, appointments, inbox, both task boards, AR, outreach. Cannot change system configuration or roles.
- **Technician** — workshop floor staff. Sees the Workshop Board (their assigned tasks), can capture photos, can update task status. Cannot see customer financial data, cannot send outreach, cannot see other technicians' boards.

Other Phase 1 capabilities:

- Immediate logout and access revocation when a staff member leaves
- Open tasks and customer assignments can be reassigned in one action during offboarding
- Audit log of all important changes (who, what, when) — basic chronological view

Future expansion (Phase 2 onwards) — full editable roles:

- KL can create custom roles (e.g., "Manager", "Apprentice Technician", "Senior Sales") with names that match how Benz Auto talks
- Per-page and per-action permission gating — KL ticks which pages each role can see and which actions they can perform
- Each workshop using the system can shape roles to their own structure (different workshops have different headcount and divisions of labour)
- Tamper-resistant audit log (append-only, downloadable as PDF or CSV)
- Searchable and filterable audit log
- Step-completion analytics linked to RBAC

Defaults Zedech has set:

- The three Phase 1 roles cover the operational reality of a small-to-medium workshop. Custom roles arrive in Phase 2.
- Superadmin role cannot be deleted (safety).
- Admin UI is in English. Customer-facing surfaces (chatbot, onboarding form) are trilingual from launch.

4.8 Accounts Receivable & Outstanding Payments (Module 12)

Purpose: Track partial settlements and follow up on debt without spreadsheets.

MVP (Phase 1):

- Manual payment entry: amount, date, method (cash / card / transfer / cheque), reference number, notes
- Automatic balance calculation: invoice total – payments received = outstanding
- Partial payments fully supported
- Debt flag on customer profile and vehicle record — visible at a glance during future bookings
- Outstanding Payments dashboard: sortable by amount, age, customer

Future expansion (Phase 2):

- Automated reminder messages for overdue accounts (via the Outreach module)
- Bulk payment recording

Defaults Zedech has set:

- No payment gateway integration. All payment recording is manual entry by staff (matches the signed scope).
- Currency: MYR, displayed in standard Malaysian format (RM 1,500.00).

4.9 Outreach Templates & Campaign Blast — MVP (Module 5)

Purpose: Send the right message to the right customer at the right time — review requests after a visit, low-rating alerts to KL, and one-off campaign blasts to customer segments.

MVP (Phase 1):

- **Template library** in English, Chinese, and Malay — admin manages all templates from the dashboard. Templates are submitted to Meta for approval
- **Manual campaign blast** — select a customer segment (e.g., dormant, high-spenders, owners of a specific make), pick a template, preview, schedule or send now
- **Collection Reminders** — remind customer about their car completion when Technician moves status to **Done**, mention to collect before office hours

- **Auto review-request** — fires automatically when an Appointment moves to **Released**. Sends template to get Google review from customer.
- **Opt-out tracking** — when a customer requests to stop, the system records it. Future messages skip them automatically
- **PDPA-compliant unsubscribe** — every outreach message includes a one-tap opt-out link
- **WhatsApp 24-hour window compliance** — outside the 24-hour customer-initiated window, the system uses Meta-approved template messages (within the window, free-form replies are allowed)
- **Personalization tokens** — customer name, last vehicle serviced, last service date
- **Automated pre-appointment** booking reminders (24 hours before, 1 hour before)

Future expansion (Phase 2 Onwards):

- RFM-segmented campaign targeting (depends on P2 Analytics — recency, frequency, monetary)
- Automated maintenance reminders (cron-based, configurable interval per service type, e.g., 6 months for general service)
- Campaign analytics: sent / delivered / read / replied counts

4.10 Digital Customer Onboarding Form (Module 13)

Purpose: A clean tablet form at the front desk so walk-in customers fill in their own details — no double-keying by your staff.

MVP (Phase 1):

- Tablet-optimized web form with large touch targets
- Fields: customer name, phone, email, preferred language, vehicle details (plate, make, model, year)
- **Trilingual from Day 1** — customer selects English, Chinese, or Malay at the top of the form
- Auto-formatting for phone numbers and license plates
- Duplicate detection: warns if phone or plate already exists in the system
- On submission, a customer profile and vehicle record are created instantly — no staff data entry

Future expansion (Phase 2):

- Optional fields configurable by admin

Defaults Zedech has set:

- The form is a URL you load on any tablet at the front desk — no app install required.
- On submission, the customer profile is auto-assigned to the next available sales staff (round-robin) until your full assignment rules go live in Phase 2.

5. Phase 2 — Workshop Operations (Weeks 9–12)

Phase 2 brings four new operational modules — Workflow Compliance, Outreach Automation, Supplier Management, AutoCount Integration, and Analytics — and layers V2 polish on top of every Phase 1 module. The Phase 1 polish includes editable roles + per-page permission gating, FB Messenger + Instagram + TikTok channels in the inbox, bay-centric Gantt calendar view, dormant flagging, auto-assignment, no-show automation, urgency flagging, multiple workshop boards, saved views, and more — see the "Future expansion" lines under each Phase 1 module for the complete list.

5.1 Enforced Workflow Compliance (Module 11)

Purpose: Make sure technicians cannot mark a task as Completed without doing the steps that matter — checklists, photos, parts verification, sign-off.

Includes in Phase 2:

- Attach mandatory steps to **any column-to-column transition you define** on either the Workshop Board or the Office Board
- Steps can include: digital inspection checklist, photo upload (before/after), parts verification, customer sign-off, voice note, plus custom steps you define
- Hard block: a card cannot move from one column to the next until all required steps for that transition are completed
- Clear error message tells the user exactly what's missing
- Mobile-friendly checklist for technicians on the workshop floor
- Step-completion analytics (which transitions slow tasks down most)
- Most common usage: attach steps to the transition *into* the Done column (e.g., "moving to Done requires photo upload + customer sign-off"). But you can attach rules to any transition — e.g., "moving from Awaiting Parts to Wrenching requires confirming parts received."

Why Phase 2: the real value of this module is the *checklists you actually want enforced* — and those are best designed after watching the workshop run with the system for 4–6 weeks during Phase 1.

Building the framework + the rules together once we've observed real workshop tasks gives you rules you'll keep, not placeholders.

Defaults Zedech has set:

- The rule applies to *everyone* including admin and manager — there is no override. This is a feature, not a bug; the whole point is that the workflow cannot be bypassed.
- The first rule will be co-designed with you and your team during a Week 9 working session: a walkthrough of one normal repair and one tricky repair, then we lock in the mandatory steps.

5.2 Outreach Automation & Segmented Campaigns — V2 (Module 5)

Purpose: Layer automation and intelligent segmentation on top of the Phase 1 outreach foundation. (Pre-appointment booking reminders already shipped in Phase 1.)

Includes in Phase 2:

- **Automated maintenance reminders** — admin-configurable interval per service type (default: 6 months for general service, 12 months for major service); sent via WhatsApp on a cron schedule
- **RFM-segmented campaign targeting** — using the segmentation produced by the Analytics module (Recency, Frequency, Monetary). Send a campaign to "high-spenders dormant for 3+ months" with one click
- **Campaign analytics** — sent / delivered / read / replied counts per campaign, per template
- **Reply-handling automation** — when a customer replies to a campaign, the message is routed to the right inbox group automatically
- **Bulk template uploads** — KL can request multiple templates in one batch; Zedech submits to Meta together

Defaults Zedech has set:

- Maintenance reminder cadences set to industry-standard intervals at launch; adjustable from the admin panel.

5.3 Supplier Management (Module 6)

Purpose: Track parts orders so nothing slips, and so late deliveries get caught before they affect the customer.

Includes in Phase 2 — manual tracking foundation:

- Supplier directory: store contacts, tags, performance score
- Three-panel layout: supplier list · conversation · order tracking & history
- **Manual order tracking** — staff create an order ticket directly from the supplier chat panel: parts list, quantities, expected delivery date, current status. Updates as the conversation progresses
- **Late-delivery alerts** — when the expected delivery date passes without a delivery confirmation, the assigned staff and admin are notified
- **Repair-date conflict alerts** — when a part is going to arrive after the scheduled appointment date for the linked technician task, surface the conflict so customer expectations can be managed early
- **Per-supplier performance dashboard** — late-delivery rate, on-time rate, average lead time

Future expansion (Phase 3) — AI augmentation:

- **AI-assisted order detection** — the system reads supplier conversations, suggests when an order has been placed, and asks a staff member to confirm with one tap (suggest-and-confirm mode)
- **AI-assisted status updates** — as the conversation continues ("delivered today", "delayed 3 days"), the AI proposes state changes for staff confirmation
- **Why Phase 3:** reliably parsing free-form, code-mixed Malaysian-English / Chinese / Malay supplier WhatsApp chats is genuinely hard work. Layering it on top of the OpenClaw platform (which installs in Phase 3) gives us the right infrastructure — strict data scoping, full audit history of every AI action, and the ability to tune on real supplier conversations captured during Phases 1–2.

Defaults Zedech has set:

- All Phase 2 alerts (late delivery, repair-date conflict) work on manually-entered orders — the Phase 3 AI layer just enriches the data source.
- Once AI is enabled in Phase 3, it runs in **suggest-and-confirm** mode for the first 2 months — staff approve each detected order with one tap. Fully autonomous detection can be turned on per-supplier later.
- Performance dashboard shows the trailing 90 days.

5.4 AutoCount Integration (Module 7)

Purpose: See live AutoCount stock and pricing from the workshop system, without re-keying.

Includes in Phase 2:

- A small bridge agent installed on the desktop machine where AutoCount runs
- View AutoCount stock levels, pricing, sales history (read-only — no changes pushed to AutoCount)
- Cached fallback when AutoCount is offline or the desktop machine is off; "last synced" timestamp shown to staff
- Manual override: when AutoCount is unavailable and a quote needs to go out, staff can override values; the system flags the override for reconciliation
- Discrepancy reconciliation queue — when AutoCount comes back online, any cached vs live mismatches surface for admin review

Defaults Zedech has set:

- Read-only access only — the workshop system never writes to AutoCount.
- Cache TTL: stock data refreshed every 15 minutes when AutoCount is online.

5.5 Sales & Margin Analytics (Module 8)

Purpose: See where the money is being made — by item, by employee, by customer.

Includes in Phase 2:

- Margin analysis: high-margin and low-margin items/services, filterable by period and category
- RFM customer segmentation (Recency, Frequency, Monetary) with sensible default thresholds and admin-adjustable bands
- Employee performance: sales volume, technician tasks completed, messages sent, average reply time
- Customer ranking by total spending, drill-down to profile
- Real-time data (not batch-processed)
- Export every report as CSV or PDF

Defaults Zedech has set:

- RFM bands use industry-standard quintiles at launch; adjustable from admin.
- Margin calculations source costs from AutoCount; if AutoCount is offline, last-known costs are used.
- Reports are timezone-aware for Malaysia (MYT).

5.6 Customer Waiting-Area Status Display

Purpose: A public-facing screen — typically a TV or large monitor mounted in the customer waiting area — showing the live status of every car currently being serviced. The same idea as the **McDonald's order-collection board**, but for cars: customers see at a glance where their vehicle is in the process. Reduces "*is my car ready yet?*" interruptions to staff and signals professionalism the moment a customer walks into the workshop.

Includes in Phase 2:

- **Bay-centric layout** — each bay is its own tile or row on the screen, showing the car currently parked there: license plate, first name only (e.g., "*John T.*"), and current status. Same visual structure as the admin-side bay-centric Gantt view, just rendered for public reading from across a room.
- **Live sync with the Workshop Board** — the status shown for each car is the Workshop Board column its technician tasks are currently sitting in. When a technician moves a card, the customer display updates within seconds. No second data entry, no risk of staff forgetting to update the display separately.
- **Status display fully customizable by admin** — KL chooses which Workshop Board columns are visible on the public screen, and what *customer-friendly label* each column gets. Internal column "Awaiting Parts" can be displayed as "*Parts arriving soon.*" "QC" can become "*Final inspection.*" "In Progress: Diagnosis" can become "*Diagnosing your car.*" Columns that aren't customer-friendly (e.g., admin-only intermediate states) can be hidden entirely.
- **Number of visible columns is up to admin** — show 3 statuses, show 7, group multiple internal columns under one public-facing label. Whatever fits Benz Auto's communication style.
- **Auto-pagination** — when more cars are in service than fit on one screen, the display cycles through pages every ~10 seconds so every customer sees their car periodically.
- **Branded** — uses the Benz Auto color palette (yellow / gray-black / silver) and large legible typography optimized for reading from waiting-area distance.
- **Privacy-aware by default** — plate + first initial of last name only (e.g., "*WMA8888 — John T.*"). Never shows phone number, full name, financial info.

How the sync works (technical, in plain language):

- The admin defines a small mapping: each Workshop Board column → either "*hide from public display*" or "*show with this customer-friendly label*."
- A car's public-display status = the customer-friendly label of the column its primary technician task is sitting in. When the technician drags the card, the public display updates near-real-time.
- When the appointment moves to **Released**, the car drops off the public display ~30 minutes later (avoids permanently showing collected cars as "Ready").

Defaults Zedech has set:

- Default mapping out of the box: any column flagged as **Done** displays as "*Ready for collection*"; everything else is hidden until KL configures customer-friendly labels.
- One public-display URL per workshop — load it on any cheap TV-stick, mini-PC, or browser. No separate app to install.
- The display is read-only — customers can't interact with it. (No risk of someone walking up and changing a status.)

5.7 Phase 1 Polish (V2 of P1 modules)

Phase 2 also delivers the "Future expansion" items listed under every Phase 1 module above. Pulled together in one place:

- **Chatbot V2:** stalled-inquiry follow-up with auto-tagging as "Potential Client"; Facebook Messenger, Instagram DM, and TikTok DM channels; jailbreak-attempt logging
- **Inbox:** Facebook Messenger, Instagram DM, and TikTok DM channels; separate "Suppliers" channel filter; advanced custom filter groups (by tags, status, channel, assignment)
- **CRM:** auto-assignment rules for new inquiries (round-robin, least-loaded, by language, by channel); dormant customer flagging
- **Appointments:** filter by bay; bay-centric Gantt view; Google Calendar one-way sync; auto-highlight Malaysian and Johor public holidays; editable holiday data; no-show automation (WhatsApp follow-up + staff task + soft profile flag)
- **Dashboard:** urgency flagging with admin-configurable delay reasons; real-time auto-refresh; touch-optimized drag-and-drop on phones; multiple Workshop Boards (e.g., one per bay or technician team); saved board views with filters
- **Access Control:** editable roles with per-page and per-action permission gating (each workshop shapes their own); tamper-resistant audit log (append-only, downloadable as PDF/CSV); searchable and filterable audit log

- **Media:** WhatsApp customer photos auto-saved to the linked vehicle (chatbot disambiguates if the customer has multiple cars); configurable retention policy with admin warning; bulk export of a vehicle's full history as PDF
- **Onboarding form:** optional fields configurable by admin
- **AR:** automated reminder messages for overdue accounts (sourced from the Outreach module); bulk payment recording

6. Phase 3 — AI Platform, Agents & Website (Weeks 13–16)

Phase 3 installs the OpenClaw AI platform, deploys the two AI agents, delivers the public Benz Auto website, and includes a final tuning pass.

6.1 OpenClaw AI Platform

Purpose: A private, secure platform on your own server where all AI assistants run, with a control panel only you can access.

Includes in Phase 3:

- Complete installation on your private Singapore VPS
- Secure admin control panel (only you have access)
- Strict data scoping — each AI assistant only sees the specific customer / appointment / task information it needs to do its work
- Official integrations connected: WhatsApp Business, Facebook Messenger, Instagram
- AI brain configuration, failover/backup setup, and AI usage cost monitoring tools
- Adjustable boundaries per AI assistant — exactly what each one is allowed to do and which parts of your business it can access
- Complete searchable history of every AI action, message, and calculation — so you can always check the work

Daily Brief

- Daily summary email at your preferred time (default 8:00 AM) in your chosen language

- Highlights: important unread messages, new sales leads, today's schedule, delayed parts, recent reviews, social media performance, new quotes, urgent items
- Same report viewable inside the app — click any item to jump to the detail
- Monday morning report includes fresh social media content ideas for the week

6.2 OpenClaw Agents

Agent #1 — Social Media & Content Manager

- Suggests weekly social media posts based on recent repairs or seasonal events
- Holds drafted posts for your review before publishing
- Auto-publishes routine posts (holiday greetings, etc.) if you enable it
- Monitors likes, comments, and DMs across your social accounts
- Summarizes social media performance each morning

Agent #2 — Pricing Strategy Agent

- Recommends pricing models — seasonal promotions, service bundles
- Models how price changes will impact profit margins
- Suggests price adjustments based on workshop capacity and peak seasons
- Develops bundled service ideas to lift average customer spend
- Reviews current parts and labor costs (from AutoCount) to recommend optimal base prices
- Monitors the success of pricing strategies over time

Note on competitor pricing: the signed quotation includes "Analyzes local market rates and competitor pricing." Reliable competitor pricing data for Johor workshops is not publicly available through any API. We propose a manual approach — you (or our team during onboarding) seed the agent with pricing data for 5–10 reference competitors, and the agent uses that as its baseline.

6.3 Customized Website (Benz Auto)

Purpose: A clean, professional public website that conveys *trust*, *reliability*, and *safety* — not the typical "wall of glossy car photos" automotive site. All "contact us" buttons funnel to WhatsApp.

Brand direction (from KL):

- **Color palette** — yellow + gray/black + silver. Silver references the Mercedes heritage; yellow + dark grays anchor the workshop identity. Used consistently across header, accents, and CTAs.
- **Tone & feel** — professional, reliable, trustworthy, sense of security. Photography and copy lean understated and substantive over flashy.
- **Hero concept** — a visual narrative of an older Mercedes gradually transforming into a newer one, communicating *"we've been focused on doing one thing well, consistently, for decades."* Could be a scroll-driven animation, a video loop, or a sequenced photo set — final treatment confirmed during design.
- **Brand story** — Benz Auto Service founded **1989**, now run by the **second generation**. This story is featured prominently on the About Us page as a trust signal, not buried.
- **Real repair case studies** — short video clips of actual repair work and customer outcomes (KL provides). These live in the Gallery and are interspersed throughout Services and Home as proof points. Authentic > polished marketing.

Pages:

- **Home** — hero (old-to-new transformation), one-line value statement, service overview, embedded case-study video, trust indicators (years in business, certifications), prominent WhatsApp CTAs
- **About Us** — brand story (1989, second-generation handover, philosophy), technician qualifications, facility and equipment showcase, service standards
- **Services** — list of all services with brief descriptions; CTAs to inquire via WhatsApp; case-study videos sprinkled in where relevant
- **Gallery** — workshop infrastructure, real repair case-study videos (primary), completed projects, customer lounge
- **Contact Us** — address with embedded map, operating hours, phone/email/messaging links, simple contact form (submits to WhatsApp or email — no separate backend)

Defaults Zedech has set:

- Mobile-first responsive design — most Malaysian car-owners discover workshops on phones
- WhatsApp floating button persistent on every page
- All CTAs use pre-filled WhatsApp message templates
- SEO optimized (meta tags, sitemap, structured data) — local Johor automotive search keywords
- Static Next.js site — fast loading, low maintenance, no backend
- Typography pairs a confident sans-serif headline face with a readable body face; minimal motion, deliberate restraint

6.4 Final Tuning (within Week 16)

The last stretch of Phase 3 is reserved for adjustments based on real-world usage of all phases — chatbot tone tuning, dashboard layout tweaks, workflow rule refinement, knowledge base seeding. This is when small things that bothered staff get fixed.

7. Maintenance, Hosting & Costs

Hosting & Security (Yearly — included in signed price)

- Domain purchase and annual renewal
- Heavy-duty web protection (WAF, DDoS protection)
- Bank-grade encryption (TLS 1.3) for all data in transit
- Zero-trust internal architecture
- Automatic secret rotation, including immediate rotation when any staff member leaves
- All databases and photos encrypted at rest
- Strict browser session security
- Multi-layer prompt-injection defense for the AI assistants

Your data isolation: Benz Auto is hosted on a dedicated Singapore VPS. Your data is not shared with any other workshop's instance.

Maintenance (Monthly — included in signed price)

- 24/7 system health monitoring with proactive alerting
- Monthly chatbot tuning based on real conversation analysis
- Monthly UI improvements based on staff feedback
- Security and dependency updates
- Background API compatibility maintenance (Meta, AutoCount, etc.)
- Routine data housekeeping (log rotation, temp file cleanup)
- AI usage included in your monthly plan (see "Ongoing usage costs" below)

Support Response Times (per signed quotation)

Severity	Response	Resolution
Critical (system down or data at risk)	30 minutes	4 hours

Severity	Response	Resolution
Major (an important feature stops working)	2 hours	24 hours
Minor (small bugs that don't stop work)	8 hours	5 business days
Small tweaks (visual changes / new ideas)	Next business day	Next work cycle

Ongoing usage costs

Two categories of usage cost arise once the system is live. Both are clarified here so there are no surprises.

Cost	Who pays	Notes
AI tokens (OpenRouter / Anthropic / OpenAI)	Zedech absorbs in normal months	Per signed quotation: standard AI usage is included; outsized busy months are billed at exact base cost with no markup. The fair-use baseline will be defined and shared with you by the end of Phase 1.
Meta WhatsApp Business outreach messages	Benz Auto pays	Meta charges per conversation (~\$0.05–0.15 USD depending on type).
Domain renewal	Zedech absorbs	Year 1 included; renewal is part of yearly hosting.
VPS	Zedech absorbs	Year 1 included in signed price.

8. How We'll Work Together

A short note on the working rhythm over the four months.

Meeting cadence — six meetings across the four months. Each phase is bracketed by two meetings: one to kick it off, one to review what shipped.

#	When	Purpose
1	Before Phase 1 (this meeting, 6 May 2026)	Walk through this PRD; lock Phase 1 scope; gather KL's inputs needed for Day 1
2	End of Phase 1 (Week 8)	Phase 1 demo + acceptance review; brief Phase 2 plan
3	Start of Phase 2 (Week 9)	Phase 2 kickoff session — confirm Phase 2 scope in detail (Workflow Compliance rules, AutoCount machine access, supplier list)
4	End of Phase 2 (Week 12)	Phase 2 demo + acceptance review; brief Phase 3 plan
5	Start of Phase 3 (Week 13)	Phase 3 kickoff — confirm OpenClaw boundaries, agent configurations, website brand assets
6	End of Phase 3 (Week 16)	Final demo + handover + tuning brief

Demos at the end of each phase walk your team through delivered modules — what works, what's defaulted, what to expect next. This is the moment for feedback before the phase is closed and the corresponding payment is due.

Feedback loop. Anything that's a fix to a delivered feature, Zedech is on the hook for and will address before the phase closes. Anything that's a new request beyond what's described in this document, Zedech will scope and quote separately so neither side is surprised.

Zedech will flag any blocker the moment it occurs.

Communication. A WhatsApp group with the project team is the default channel for day-to-day questions and updates. Weekly check-ins to keep alignment.

This document is a shared description of the project, prepared for discussion. If anything here doesn't match your understanding, please flag it and we'll adjust.

Phase 1 Acknowledgement

By signing below, I confirm that I have read and reviewed this document and agree that it reflects the scope, defaults, and approach for **Phase 1** of the Benz Auto Workshop Management System project. Phase 2 and Phase 3 will have their own kickoff meetings and acknowledgements at the appropriate time (per the meeting cadence above).

This acknowledgement does not modify the commercial terms of the quotation dated 15 April 2026 (signed 5 May 2026).

Acknowledged for Benz Auto Service – M Sdn Bhd:

Signature: _____

Name: KL Chan
Benz Auto Service (M) Sdn Bhd

Date: 7 May 2026

Acknowledged for Zedech Solution:

Signature: _____

Name: Tay Jia Chenz
Zedech Solution

Date: 7 May 2026